

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Inverse Sine with Exponential

Evaluate the definite integral:

$$\int_0^1 x(x+3)e^x \sin^{-1}(x) dx$$

Solution: We use integration by parts. Let $u = \sin^{-1}(x)$ and $dv = (x^2 + 3x)e^x dx$. To find v , we perform tabular integration on dv :

$$\int (x^2 + 3x)e^x dx = (x^2 + 3x)e^x - \int (2x + 3)e^x dx = (x^2 + 3x)e^x - (2x + 3)e^x + 2e^x = (x^2 + x - 1)e^x$$

Thus, $v = (x^2 + x - 1)e^x$. Applying the integration by parts formula $\int u dv = uv - \int v du$:

$$I = \left[(x^2 + x - 1)e^x \sin^{-1}(x) \right]_0^1 - \int_0^1 \frac{(x^2 + x - 1)e^x}{\sqrt{1 - x^2}} dx$$

Evaluate the boundary term:

$$\left[(1^2 + 1 - 1)e^1 \sin^{-1}(1) \right] - \left[(-1)e^0 \sin^{-1}(0) \right] = e \left(\frac{\pi}{2} \right) - 0 = \frac{\pi e}{2}$$

Now, we analyze the remaining integral. We can rewrite the numerator algebraically:

$$x^2 + x - 1 = -(1 - x^2) + x$$

This clever decomposition splits the integrand into two manageable terms:

$$\int_0^1 \frac{-(1 - x^2) + x}{\sqrt{1 - x^2}} e^x dx = \int_0^1 \left(-\sqrt{1 - x^2} + \frac{x}{\sqrt{1 - x^2}} \right) e^x dx$$

Notice that the integrand perfectly matches the product rule expansion of the derivative of $-\sqrt{1 - x^2}e^x$:

$$\frac{d}{dx} \left(-\sqrt{1 - x^2}e^x \right) = \frac{x}{\sqrt{1 - x^2}}e^x - \sqrt{1 - x^2}e^x$$

Hence, the integral evaluates trivially by the Fundamental Theorem of Calculus:

$$\int_0^1 \frac{d}{dx} \left(-\sqrt{1 - x^2}e^x \right) dx = \left[-\sqrt{1 - x^2}e^x \right]_0^1 = (-0 \cdot e) - (-1 \cdot 1) = 1$$

Subtracting this from the evaluated boundary term yields the final exact answer:

$$I = \frac{\pi e}{2} - 1$$

1.2 Hard Integral

Error Function and Exponential

Evaluate the definite integral:

$$\int_0^{\infty} \frac{\operatorname{erf}(\sqrt{x})}{xe^x} dx$$

Solution: We introduce a parameter a to utilize Feynman's trick. Let:

$$I(a) = \int_0^{\infty} \frac{\operatorname{erf}(\sqrt{ax})}{xe^x} dx$$

We wish to evaluate $I(1)$. Note that $I(0) = 0$. Differentiate $I(a)$ with respect to a . The derivative of the error function is $\frac{d}{da} \operatorname{erf}(\sqrt{ax}) = \frac{1}{\sqrt{\pi}} \frac{\sqrt{x}}{\sqrt{a}} e^{-ax}$.

$$I'(a) = \int_0^{\infty} \frac{1}{xe^x} \left(\frac{1}{\sqrt{\pi}} \frac{\sqrt{x}}{\sqrt{a}} e^{-ax} \right) dx = \frac{1}{\sqrt{\pi a}} \int_0^{\infty} x^{-1/2} e^{-(a+1)x} dx$$

Apply the substitution $u = (a+1)x$, which gives $dx = \frac{du}{a+1}$ and $x^{-1/2} = \sqrt{a+1} u^{-1/2}$:

$$I'(a) = \frac{1}{\sqrt{\pi a}} \frac{1}{\sqrt{a+1}} \int_0^{\infty} u^{-1/2} e^{-u} du$$

Recognizing the gamma function integral $\int_0^{\infty} u^{-1/2} e^{-u} du = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, we obtain:

$$I'(a) = \frac{1}{\sqrt{a(a+1)}}$$

Integrate $I'(a)$ from $a = 0$ to $a = 1$ to recover $I(1)$:

$$I(1) - I(0) = \int_0^1 \frac{1}{\sqrt{a^2 + a}} da$$

Using the standard antiderivative $\int \frac{du}{\sqrt{u^2 + u}} = \ln \left| u + \frac{1}{2} + \sqrt{u^2 + u} \right|$:

$$I(1) = \left[\ln \left(a + \frac{1}{2} + \sqrt{a^2 + a} \right) \right]_0^1$$

Evaluating at the limits:

$$I(1) = \ln \left(1 + \frac{1}{2} + \sqrt{2} \right) - \ln \left(\frac{1}{2} + 0 \right) = \ln \left(\frac{3 + 2\sqrt{2}}{2} \right) - \ln \left(\frac{1}{2} \right)$$

$$I = \ln(3 + 2\sqrt{2}) = 2\ln(1 + \sqrt{2})$$

Differentials

2.1 Medium Differential

Invertible First-Order ODE

Consider the differential equation:

$$\frac{dy}{dx} = \frac{y}{2 - x + \ln(y)}$$

Given that the solution curve passes through the point $(1, 1)$, find the exact value of x when $y = 87$.

Solution: The ODE is highly non-linear in y , but can be simplified by treating x as a function of y . Take the reciprocal of both sides:

$$\frac{dx}{dy} = \frac{2 - x + \ln(y)}{y} = \frac{2 + \ln(y)}{y} - \frac{1}{y}x$$

Rearranging yields a standard first-order linear differential equation:

$$\frac{dx}{dy} + \frac{1}{y}x = \frac{2 + \ln(y)}{y}$$

The integrating factor (IF) is:

$$\mu(y) = \exp\left(\int \frac{1}{y} dy\right) = \exp(\ln(y)) = y$$

Multiplying the entire equation by the IF gives:

$$y \frac{dx}{dy} + x = 2 + \ln(y) \implies \frac{d}{dy}(xy) = 2 + \ln(y)$$

Integrate both sides with respect to y :

$$xy = \int (2 + \ln(y)) dy = 2y + (y \ln(y) - y) + C = y + y \ln(y) + C$$

Dividing through by y , we get the general solution:

$$x(y) = 1 + \ln(y) + \frac{C}{y}$$

Apply the initial condition that the curve passes through $(1, 1)$:

$$1 = 1 + \ln(1) + \frac{C}{1} \implies 1 = 1 + 0 + C \implies C = 0$$

The specific solution is $x(y) = 1 + \ln(y)$. To find x when $y = 87$, we simply evaluate:

$$x(87) = 1 + \ln(87)$$

2.2 Hard Differential

Second-Order Non-Linear ODE

A twice-differentiable non-constant function $y = f(x)$ satisfies the second-order differential equation:

$$y'' + 3yy' + y^3 = 0$$

Given $f(0) = 0$ and $f'(0) = 2$, find the exact value of:

$$\int_0^1 y(x) dx$$

Solution: We apply the Riccati-style substitution $y = \frac{u'}{u}$. This gives:

$$y' = \frac{u''}{u} - \frac{(u')^2}{u^2}$$

$$y'' = \frac{u'''}{u} - \frac{3u'u''}{u^2} + \frac{2(u')^3}{u^3}$$

Substitute these expressions into the original ODE:

$$\left(\frac{u'''}{u} - \frac{3u'u''}{u^2} + \frac{2(u')^3}{u^3} \right) + 3 \left(\frac{u'}{u} \right) \left(\frac{u''}{u} - \frac{(u')^2}{u^2} \right) + \left(\frac{u'}{u} \right)^3 = 0$$

Expanding and carefully canceling the terms:

$$\frac{u'''}{u} - \frac{3u'u''}{u^2} + \frac{2(u')^3}{u^3} + \frac{3u'u''}{u^2} - \frac{3(u')^3}{u^3} + \frac{(u')^3}{u^3} = 0$$

Remarkably, all non-linear terms cancel out perfectly, leaving:

$$\frac{u'''}{u} = 0 \implies u''' = 0$$

Integrating three times, we find $u(x) = Ax^2 + Bx + C$. Returning to y :

$$y(x) = \frac{2Ax + B}{Ax^2 + Bx + C}$$

Using the initial condition $f(0) = 0$:

$$y(0) = \frac{B}{C} = 0 \implies B = 0$$

This simplifies the function to $y(x) = \frac{2Ax}{Ax^2 + C}$. Now, find $y'(x)$ to apply the second initial condition $f'(0) = 2$:

$$y'(x) = \frac{2A(Ax^2 + C) - 2Ax(2Ax)}{(Ax^2 + C)^2} = \frac{2AC - 2A^2x^2}{(Ax^2 + C)^2}$$

Evaluating at $x = 0$:

$$y'(0) = \frac{2AC}{C^2} = \frac{2A}{C} = 2 \implies 2A = 2C \implies A = C$$

Substituting $A = C$ back into $y(x)$:

$$y(x) = \frac{2Cx}{Cx^2 + C} = \frac{2x}{x^2 + 1}$$

Finally, evaluate the definite integral:

$$\int_0^1 \frac{2x}{x^2 + 1} dx = \left[\ln(x^2 + 1) \right]_0^1 = \ln(2) - \ln(1) = \ln(2)$$

Derivatives

3.1 Medium Derivative

Infinite Nested Radical

Let $f(x) = \sqrt{-4x^2 + 4x \left(\sqrt{-4x^2 + 4x(\sqrt{\dots})} \right)}$. Find $f'(x)$.

Solution: Let $y = f(x)$. Due to the infinite nature of the nested radical, the expression inside the parenthesis is identically y itself. Thus, we can rewrite the equation as:

$$y = \sqrt{-4x^2 + 4xy}$$

Squaring both sides to eliminate the outer radical yields:

$$y^2 = -4x^2 + 4xy$$

Rearranging the terms into a standard polynomial form gives a perfect square trinomial:

$$\begin{aligned} y^2 - 4xy + 4x^2 &= 0 \\ (y - 2x)^2 &= 0 \end{aligned}$$

Taking the square root of both sides gives a simple linear relationship:

$$y = 2x$$

Therefore, the function simplifies drastically to $f(x) = 2x$. Taking the derivative with respect to x is straightforward:

$$f'(x) = 2$$

3.2 Hard Derivative

Iterated Rational Function

Let $f(x) = 1 - \frac{1}{x}$ and $g(x) = \underbrace{f(f(\dots f(x) \dots))}_{2026 \text{ times}}$. Find $g'(f'(\pi))$.

Solution: First, we analyze the behavior of the iterates of $f(x)$ to find a pattern.

$$\begin{aligned} f^1(x) &= 1 - \frac{1}{x} = \frac{x-1}{x} \\ f^2(x) &= f(f(x)) = 1 - \frac{1}{\frac{x-1}{x}} = 1 - \frac{x}{x-1} = \frac{x-1-x}{x-1} = \frac{-1}{x-1} = \frac{1}{1-x} \\ f^3(x) &= f(f^2(x)) = 1 - \frac{1}{\frac{1}{1-x}} = 1 - (1-x) = x \end{aligned}$$

Since $f^3(x) = x$, the sequence of functions is periodic with a period of 3. We determine the position of the 2026th iterate within this cycle:

$$2026 = 3 \times 675 + 1 \implies 2026 \equiv 1 \pmod{3}$$

Consequently, the 2026th iteration is identical to the 1st iteration:

$$g(x) = f^{2026}(x) = f^1(x) = 1 - \frac{1}{x}$$

We need to find the value of $g'(f'(\pi))$. First, find $f'(x)$:

$$f'(x) = \frac{d}{dx} \left(1 - x^{-1} \right) = x^{-2} = \frac{1}{x^2}$$

Evaluate this at $x = \pi$:

$$f'(\pi) = \frac{1}{\pi^2}$$

Next, find $g'(x)$. Since $g(x) = f(x)$, their derivatives are identical:

$$g'(x) = \frac{1}{x^2}$$

Finally, evaluate $g'(x)$ at the input value $f'(\pi)$:

$$g'(f'(\pi)) = g' \left(\frac{1}{\pi^2} \right) = \frac{1}{\left(\frac{1}{\pi^2} \right)^2} = \frac{1}{\frac{1}{\pi^4}}$$

$$g'(f'(\pi)) = \pi^4$$

Physics & Engineering

4.1 Mechanics

Magnetic Braking of a Sliding Rod

A long horizontal conductive rod of length $L = 1.5$ m is located in a uniform magnetic field $B = 69$ mT, perpendicular to the plane of the figure. A conductive jumper with mass $m = 6.7$ kg and friction coefficient $\mu = 0.21$ slides along the rod, completing a circuit with resistance $R = 9 \Omega$. At time $t = 0$, the jumper is given an initial velocity $v_0 = 4$ m s⁻¹. Find the maximum stopping distance. Take $g = 9.81$ m s⁻².

Solution: The jumper experiences two opposing forces: kinetic friction $F_f = \mu mg$ and the induced magnetic Lorentz force $F_B = ILB$. The induced electromotive force is $\mathcal{E} = BLv$, yielding an induced current $I = \frac{BLv}{R}$. Newton's Second Law for the jumper gives:

$$m \frac{dv}{dt} = -\frac{B^2 L^2 v}{R} - \mu mg$$

Using the chain rule $\frac{dv}{dt} = v \frac{dv}{dx}$, we separate variables to find the stopping distance x_{max} :

$$mv \frac{dv}{dx} = -\left(\frac{B^2 L^2}{R} v + \mu mg\right) \implies dx = \frac{-mv}{\frac{B^2 L^2}{R} v + \mu mg} dv$$

Let $k = \frac{B^2 L^2}{R}$. Integrating from initial velocity v_0 to 0:

$$x_{max} = \int_{v_0}^0 \frac{-mv}{kv + \mu mg} dv = \int_0^{v_0} \frac{mv}{kv + \mu mg} dv$$

Performing algebraic division on the integrand:

$$x_{max} = \frac{m}{k} \int_0^{v_0} \left(1 - \frac{\mu mg}{kv + \mu mg}\right) dv = \frac{m}{k} \left[v_0 - \frac{\mu mg}{k} \ln \left(1 + \frac{kv_0}{\mu mg}\right) \right]$$

Now, calculate the constants:

$$k = \frac{(0.069)^2 (1.5)^2}{9} = \frac{0.004761 \times 2.25}{9} \approx 0.00119 \text{ kg s}^{-1}$$

$$\mu mg = 0.21 \times 6.7 \times 9.81 \approx 13.80 \text{ N}$$

Since $kv_0 \ll \mu mg$, the magnetic damping force is exceptionally small relative to friction. We can approximate the logarithm using the Taylor expansion $\ln(1 + u) \approx u - \frac{u^2}{2}$:

$$x_{max} \approx \frac{m}{k} \left[v_0 - \frac{\mu mg}{k} \left(\frac{kv_0}{\mu mg} - \frac{1}{2} \left(\frac{kv_0}{\mu mg} \right)^2 \right) \right] = \frac{mv_0^2}{2\mu mg}$$

Substitute the values into the dominant frictional stopping distance term:

$$x_{max} \approx \frac{4^2}{2 \times 0.21 \times 9.81} = \frac{16}{4.1202} \approx 3.88 \text{ m}$$

4.2 Quantum Mechanics

Electron Diffraction

A narrow beam of electrons with a kinetic energy of $K = 15.0$ keV is directed through a polycrystalline aluminium foil, producing a diffraction pattern on a screen at a distance $\ell = 12.0$ cm. A specific diffraction ring of diameter $D = 7.0$ cm is observed. Determine the interplanar distance d (in pm) of the crystal planes responsible for this ring, assuming it corresponds to a third-order reflection ($n = 3$). Assume $h = 6.0 \times 10^{-34}$ J s, $m_e = 9.0 \times 10^{-31}$ kg, and $e = 1.5 \times 10^{-19}$ C.

Solution: First, calculate the non-relativistic momentum of the electrons. The given charge unit $e = 1.5 \times 10^{-19}$ C yields the kinetic energy in Joules:

$$K = 15.0 \times 10^3 \text{ eV} \times (1.5 \times 10^{-19} \text{ J/eV}) = 22.5 \times 10^{-16} \text{ J}$$

The electron momentum p is given by:

$$p = \sqrt{2m_e K} = \sqrt{2 \times (9.0 \times 10^{-31}) \times (22.5 \times 10^{-16})} = \sqrt{405 \times 10^{-47}} = \sqrt{40.5} \times 10^{-23} \text{ kg m s}^{-1}$$

Notice that $\sqrt{40.5} = \sqrt{\frac{81}{2}} = \frac{9}{\sqrt{2}}$. Next, find the de Broglie wavelength λ :

$$\lambda = \frac{h}{p} = \frac{6.0 \times 10^{-34}}{\frac{9}{\sqrt{2}} \times 10^{-23}} = \frac{6\sqrt{2}}{9} \times 10^{-11} \text{ m} = \frac{2\sqrt{2}}{3} \times 10^{-11} \text{ m} = \frac{20\sqrt{2}}{3} \text{ pm}$$

Now, determine the scattering angle θ from the geometry. The ring radius is $r = D/2 = 3.5$ cm. The relation to the screen distance is:

$$\tan(2\theta) = \frac{r}{\ell} = \frac{3.5}{12.0} = \frac{7}{24}$$

In a right triangle with legs 7 and 24, the hypotenuse is 25. Thus, $\cos(2\theta) = \frac{24}{25}$. Using the half-angle identity for sine:

$$\sin(\theta) = \sqrt{\frac{1 - \cos(2\theta)}{2}} = \sqrt{\frac{1 - 24/25}{2}} = \sqrt{\frac{1/25}{2}} = \frac{1}{\sqrt{50}} = \frac{\sqrt{2}}{10}$$

Finally, apply Bragg's Law $2d \sin(\theta) = n\lambda$ for the third-order reflection ($n = 3$):

$$2d \left(\frac{\sqrt{2}}{10} \right) = 3 \left(\frac{20\sqrt{2}}{3} \right)$$

Simplifying both sides:

$$d \frac{\sqrt{2}}{5} = 20\sqrt{2} \implies d = 20 \times 5 = 100 \text{ pm}$$

4.3 Electromagnetism

Interaction Force of a Charged Ring and Thread

A thin charged wire ring has radius $R = 30$ cm and total charge $q = 2.0 \mu\text{C}$. A very long uniformly charged thread with linear charge density $\lambda = 5.0 \mu\text{C/m}$ is placed along the axis of the ring, with one end at the center of the ring. Find the magnitude of the interaction force F (in N) between the ring and the thread. Take $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$.

Solution: By Newton's Third Law, the force exerted by the thread on the ring is equal in magnitude to the force exerted by the ring on the thread. It is simpler to calculate the latter. The electric field $E(z)$ produced by a charged ring at a distance z along its central axis is:

$$E(z) = \frac{kqz}{(R^2 + z^2)^{3/2}}$$

The infinitesimal force dF on an infinitesimal segment dz of the charged thread (which carries charge $dq = \lambda dz$) is:

$$dF = E(z)dq = \frac{kq\lambda z}{(R^2 + z^2)^{3/2}} dz$$

To find the total interaction force F , integrate dF from $z = 0$ (the center of the ring) to $z = \infty$ (the end of the long thread):

$$F = \int_0^\infty \frac{kq\lambda z}{(R^2 + z^2)^{3/2}} dz$$

We use the standard power rule integration, recognizing the derivative of the inner function $R^2 + z^2$ is $2z$:

$$F = kq\lambda \left[-\frac{1}{\sqrt{R^2 + z^2}} \right]_0^\infty$$

Evaluating this at the upper and lower bounds:

$$F = kq\lambda \left(0 - \left(-\frac{1}{\sqrt{R^2 + 0}} \right) \right) = \frac{kq\lambda}{R}$$

Substitute the numerical values provided:

$$F = \frac{(9 \times 10^9) \times (2.0 \times 10^{-6}) \times (5.0 \times 10^{-6})}{0.30}$$

$$F = \frac{90 \times 10^{-3}}{0.30} = 300 \times 10^{-3} \text{ N} = 0.3 \text{ N}$$